

SYNOPSIS

Product Sentiment Analysis and Feature Roadmap Prioritization Using Python

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1. Problem Statement

1.1 Title

Product Sentiment Analysis and Feature Roadmap Prioritization Using Python

1.2 Background and Context

Mobile applications generate thousands of user reviews that provide valuable insights into customer satisfaction, feature requests, and product issues. However, manually analyzing large volumes of reviews is inefficient and time-consuming. This project focuses on the **FitLife Health and Fitness** application, which faces challenges such as declining user retention, mixed customer feedback, and limited engineering resources. The objective is to use Natural Language Processing (NLP) and sentiment analysis to extract meaningful insights and support data-driven product decisions.

1.3 Who Is Affected and Why It Matters

Product managers, developers, business teams, and end users are directly affected by inefficient feature prioritization. A data-driven sentiment analysis system enables organizations to understand customer needs, improve user satisfaction, allocate development resources effectively, and enhance overall product quality.

1.4 Objectives

- Classify user reviews into Positive, Negative, and Neutral sentiments.
- Extract important features and customer pain points using NLP techniques.
- Prioritize product features using the RICE (Reach, Impact, Confidence, Effort) framework.
- Develop dashboards and business recommendations for product improvement.

1.5 Scope and Limitations

Scope

- Analyze Health & Fitness app reviews from the Google Play Store dataset.
- Perform text preprocessing, sentiment analysis, feature extraction, and RICE prioritization.
- Generate visual dashboards and actionable product insights.

Limitations

- Results depend on dataset quality and review authenticity.
- Sentiment analysis may misclassify sarcastic or ambiguous reviews.
- RICE scoring is based on estimated business assumptions.
- The study is limited to Health & Fitness mobile applications.

2. Method Used (Methodology)

2.1 Proposed Approach

Data Collection → Data Cleaning → Text Preprocessing → Sentiment Analysis (VADER) → Feature Extraction (TF-IDF & NLP) → RICE Prioritization → Dashboard & Business Recommendations

2.2 Python Libraries and Tools

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn and MS-Office are used for data preprocessing, sentiment analysis, feature extraction, visualization, and dashboard development.

2.3 Input Data Source and Data Flow

Data Import → Data Cleaning → Text Preprocessing → Sentiment Analysis → Feature Extraction → RICE Prioritization → Dashboard

2.4 System / Module Design

#	Module	Function
1	Data Import	Load review dataset
2	Data Cleaning	Handle missing values and duplicates.
3	Text Preprocessing	Tokenization, stop-word removal, stemming
4	Sentiment Analysis	Classify customer reviews using VADER
5	Feature Extraction	Identify keywords and feature requests using TF-IDF
6	Feature Prioritization	Rank features using the RICE framework
7	Dashboard	Display insights and business recommendations

Block Diagram:

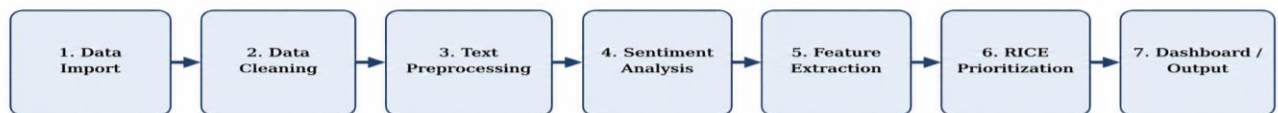


Figure 1: Module Flow

2.5 Expected Output and Evaluation Approach

Expected Output

- Sentiment classification of customer reviews.
- Identification of key feature requests and pain points.
- Feature prioritization using RICE scores.
- Interactive dashboard with visualizations and recommendations.

Evaluation Approach

The system will be evaluated based on sentiment classification accuracy, correctness of feature extraction, effectiveness of RICE prioritization, dashboard quality, and the usefulness of recommendations for product managers.